More about vertex connectivity of special classes of divisible design graphs

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Deza graphs

Deza graphs were introduced in 1999 [1] as a generalisation of strongly regular graphs.

A k-regular graph on v vertices is called a Deza graph with parameters (v, k, b, a), $b \geq a$ if the number of common neighbours of any two distinct vertices takes two values: a or b.

A Deza graph is called a strictly Deza graph if it has diameter 2 and is not strongly regular.

[1] M. Erickson, S. Fernando, W.H. Haemers, D. Hardy, J. Hemmeter, Deza graphs: A generalization of strongly regular graphs, *J. Comb. Des.*, **7** no. 6, (1999) 359–405

Divisible design graphs

A k-regular graph on v vertices is called a divisible design graph (DDG for short) with parameters $(v, k, \lambda_1, \lambda_2, m, n)$ if its vertex set can be partitioned into m classes of size n, such that two distinct vertices from the same class have exactly λ_1 common neighbors, and two vertices from different classes have exactly λ_2 common neighbors.

The definition implies that divisible design graphs are Deza graphs.

Feasible parameters of divisible design graphs

Divisible design graphs were first studied in master's thesis by M.A. Meulenberg [2].

In three following papers [3, 4, 5] DDGs were studied in more details and a lot of new constructions were presented.

[2] M.A. Meulenberg, Divisible Design Graphs, Master's thesis, *Tilburg University*, 2008.

[3] W.H. Haemers, H. Kharaghani, M.A. Meulenberg, Divisible Design Graphs, *J. of Comb. Th., Series A*, **118** (2011) 978–992

[4] D. Crnkovic, W.H. Haemer, More about Divisible Design Graphs, *CentER Discussion Paper*, **2011-140** (2011)

[5] D. Crnkovic, W.H. Haemer, Walk-regular divisible design graphs.Designs,*Codes and Cryptography*, **72** (2014) 165–175

Eigenvalues of divisible design graphs

Divisible Design Graphs graphs have at most 5 distinct eigenvalues. Formulas for eigenvalues and their multiplicities are presented in the table:

Eigenvalues	Multiplicity	
$\theta_0 = k$	1	
$\theta_1 = \sqrt{k-\lambda_1}$	f_1	
$\theta_2 = -\sqrt{k-\lambda_1}$	f_2	$f_1 + f_2 = m(n-1)$
$\theta_3 = \sqrt{k^2-\lambda_2 v}$	g_1	
$\theta_4 = -\sqrt{k^2-\lambda_2 v}$	g_2	$g_1 + g_2 = m - 1$

Vertex connectivity

The vertex connectivity $\varkappa(\Gamma)$ of the graph Γ is the minimum number of vertices whose deletion from Γ disconnects it.
For example, the vertex connectivity of a strongly regular graph is equal to its valency, which was proved by Brouwer and Mesner in 1985 ([6]).
A similar statement for distance-regular graphs was proved by Brouwer and Koolen in 2009 ([7]).

[6] A. E. Brouwer and D. M. Mesner, The connectivity of strongly regular graphs, Europ. J. Combin. **6**, 215–216 (1985).
[7] A. E. Brouwer and J. H. Koolen, The vertex-connectivity of a distance-regular graph, Europ. J. Combin. **30** (3), 668–673 (2009).

Vertex connectivity

Let x and y be two different vertices of a graph Γ. Two simple paths connecting x and y are called disjoint if they have no common vertices different from x and y.
A set of vertices S disconnects x and y if x and y belong to different connected components of the graph $\Gamma \setminus S$.
A set S of vertices of a graph Γ is called disconnecting if it disconnects some two of its vertices.

Vertex connectivity of Deza graphs

Theorem 1 ([8, 9])

Let Δ be a Deza graph obtained from a strongly regular graph Γ with restricted eigenvalues $r > 0$ and $s < 0$ by dual Seidel switching. Then, we have one of the following two cases:
(1) if $r \neq 2$, the vertex connectivity of Δ equals to its valency, except for the case when Γ is a complement of $n \times n$-lattice, in that case the vertex connectivity is one less then valency;
(2) $r = 2$.

[8] A.L. Gavrilyuk, S.V. Goryainov, V.V. Kabanov, On the vertex connectivity of Deza graphs, Proc. Steklov Inst. Math. 285 (Suppl 1) (2014) 68–77.
[9] S.V. Goryainov, D.I. Panasenko, On vertex connectivity of Deza graphs with parameters of the complements to Seidel graphs, *European Journal of Combinatorics* **80** (2019) 143–150.

Previous results

Lemma 1

The vertex connectivity of a connected DDG Γ with parameters $(v, k, \lambda_1, \lambda_2, m, n)$, where $\lambda_2 = 0$, equals k.

Lemma 2

The vertex connectivity of a connected DDG Γ with parameters $(v, k, \lambda_1, \lambda_2, m, n)$, where $\lambda_1 = k$, equals k.

Previous results

Lemma 3

Let Γ be a DDG with parameters $(v, k, \lambda_1, \lambda_2, m, n)$, where $\lambda_2 = 2k - v$. Then the following statements hold.
(1) If $\lambda_1 \neq k - 1$, then the vertex connectivity of Γ equals k.
(2) If $\lambda_1 = k - 1$, then the vertex connectivity of Γ equals $k - 1$.

Lemma 4

The vertex connectivity of a DDG Γ with parameters $(v, k, \lambda_1, \lambda_2, m, n)$, where $\lambda_1 = k - 1$ and $\lambda_2 \notin \{0, 2k - v\}$, equals $k - 1$.

Hadamard matrices

An $m \times m$ matrix H is a Hadamard matrix if every entry is 1 or -1 and $HH^\top = mI$. A Hadamard matrix H is called graphical if H is symmetric with constant diagonal, and regular if all row and column sums are equal.

Construction 1 ([3, Construction 4.8]

Consider a regular graphical Hadamard matrix H of order $l^2 \geqslant 4$ with diagonal entries -1 and row sum l. The graph with adjacency matrix

$$A = \begin{bmatrix} M & N & O \\ N & O & M \\ O & M & N \end{bmatrix},$$

where

$$M = \frac{1}{2}\begin{bmatrix} J+H & J+H \\ J+H & J+H \end{bmatrix} \text{ and } N = \frac{1}{2}\begin{bmatrix} J+H & J-H \\ J-H & J+H \end{bmatrix},$$

is a DDG with parameters $(6l^2,\ 2l^2+l,\ l^2+l,\ (l^2+l)/2,\ 3,\ 2l^2)$.

Vertex connectivity of DDGs

Let Γ be a DDG with parameters $(6l^2,\, 2l^2+l,\, l^2+l,\, (l^2+l)/2,\, 3,\, 2l^2)$ obtained with Construction 1 with positive l.
Consider the subgraph induced by the first $2l^2$ vertices of Γ (in terms of Construction 1 this subgraph has adjacency matrix M) and the subgraph induced by the last $2l^2$ vertices of Γ (in terms of Construction 1 this subgraph has adjacency matrix N).
Denote by Γ_1 and Γ_2 the first and the second subgraph, respectively.

Vertex connectivity of DDGs

Lemma 5

Γ_1 is an $(l^2 + l)$-regular graph and Γ_2 is an l^2-regular graph.

Lemma 6

The vertex connectivity of Γ is at most $2l^2$.

Lemma 7

If $\varkappa(\Gamma_1) \geq l^2, \varkappa(\Gamma_2) \geq l^2 - l$ and $\varkappa(\Gamma_1) + \varkappa(\Gamma_2) \geq 2l^2$, then the vertex connectivity of Γ equals $2l^2$.

Sketch of proof:

Denote by Γ_0 the coclique induced by the $2l^2$ vertices of Γ that correspond to the middle rows and columns of A. Note that any vertex from Γ_1 has exactly l^2 neighbours in Γ_0 and any vertex from Γ_2 has exactly $l^2 + l$ neighbours in Γ_0.

Let t_1, t_2 and t_0 be non-negative integers, such that $t_1 + t_2 + t_0 < 2l^2$. If $t_1 \geq \varkappa(\Gamma_1)$ and $t_2 \geq \varkappa(\Gamma_2)$, then $t_1 + t_2 \geq 2l^2$, which contradicts to the inequality above. Denote by $\hat{\Gamma}_1, \hat{\Gamma}_2$ and $\hat{\Gamma}_0$ the graphs obtained by deletion of t_1, t_2 and t_0 vertices from Γ_1, Γ_2 and Γ_0, respectively (also denote by $\hat{\Gamma}$ the graph obtained by deletion of $t_1 + t_2 + t_0$ vertices from Γ).

Consider three possible cases:
Case 1: $t_1 < \varkappa(\Gamma_1)$ and $t_2 < \varkappa(\Gamma_2)$. Then $\hat{\Gamma}_1$ and $\hat{\Gamma}_2$ are connected. Thus, if there is at least one vertex in $\hat{\Gamma}_0$ that has neighbours in both $\hat{\Gamma}_1$ and $\hat{\Gamma}_2$, then $\hat{\Gamma}$ is connected. Let us show that such a vertex exists.
Since Γ_2 is an l^2-regular graph, we have the inequality $\varkappa(\Gamma_2) \le l^2$ and, consequently, $t_2 < l^2$. Since l^2 is less than $l^2 + l$, which is the number of neighbours in Γ_0 for any vertex from Γ_2, we conclude that any vertex from Γ_0 is adjacent to at least one vertex from $\hat{\Gamma}_2$. This means that any vertex from $\hat{\Gamma}_0$ is adjacent to at least one vertex from $\hat{\Gamma}_2$.

Vertex connectivity of DDGs

Consider two subcases: $t_1 < l^2$ and $t_1 \geq l^2$.
If $t_1 < l^2$, an argument similar to the argument above implies that any vertex from $\hat{\Gamma}_0$ is adjacent to at least one vertex from $\hat{\Gamma}_1$. So there exists a vertex from $\hat{\Gamma}_0$ that has neighbours in both $\hat{\Gamma}_1$ and $\hat{\Gamma}_2$.
If $t_1 \geq l^2$, then the inequalities $t_1 + t_2 + t_0 < 2l^2$ and $t_2 \geq 0$ imply that t_0 is less than l^2, which is the number of neighbours in Γ_0 for any vertex from Γ_1. Therefore any vertex from $\hat{\Gamma}_1$ is adjacent to at least one vertex from $\hat{\Gamma}_0$. So there exists a vertex from $\hat{\Gamma}_0$ that has neighbours in both $\hat{\Gamma}_1$ and $\hat{\Gamma}_2$.

Case 2: $t_1 \geq \varkappa(\Gamma_1)$ and $t_2 < \varkappa(\Gamma_2)$. Then $\hat{\Gamma}_1$ is disconnected and $\hat{\Gamma}_2$ is connected. Thus, if, for each connected component of $\hat{\Gamma}_1$, there is at least one vertex in $\hat{\Gamma}_0$ having neighbours in both this component and $\hat{\Gamma}_2$, then $\hat{\Gamma}$ is connected. Let us show that such vertices exist.
Since $t_2 < \varkappa(\Gamma_2) \leq l^2$, any vertex from $\hat{\Gamma}_0$ is adjacent to at least one vertex from $\hat{\Gamma}_2$ (see Case 1).
Since $t_1 \geq \varkappa(\Gamma_1) \geq l^2$ and $t_2 \geq 0$, the inequality $t_0 < l^2$ holds. So any vertex from $\hat{\Gamma}_1$ is adjacent to at least one vertex from $\hat{\Gamma}_0$ (see Case 1).
So, for each connected component of $\hat{\Gamma}_1$, there is a vertex from $\hat{\Gamma}_0$ having neighbours in both this component and $\hat{\Gamma}_2$.
Case 3: $t_1 < \varkappa(\Gamma_1)$ and $t_2 \geq \varkappa(\Gamma_2)$. This case is similar to Case 2.

Hadamard matrices

If H_1 and H_2 are Hadamard matrices, then so is the Kronecker product $H_1 \otimes H_2$. Moreover, if H_1 and H_2 are regular with row sums l_1 and l_2, respectively, then $H_1 \otimes H_2$ is regular with row sum $l_1 l_2$. Similarly, the Kronecker product of two graphical Hadamard matrices is graphical again.

Consider regular graphical Hadamard matrices H and H', where

$$H = \begin{bmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix} \text{ and } H' = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{bmatrix}.$$

Denote by H_1 the matrix H. For any integer t such that $t > 1$, denote by H_t the Kronecker product $H_{t-1} \otimes H'$. The matrix H_t is a regular graphical Hadamard matrices of order 4^t with diagonal entries -1 and row sum 2^t.

Applying Construction 1 to H_t, we obtain a DDG with parameters $(6 \cdot 4^t,\ 2 \cdot 4^t + 2^t,\ 4^t + 2^t,\ 2 \cdot 4^{t-1} + 2^{t-1},\ 3,\ 2 \cdot 4^t)$. The smallest example is a DDG with parameters $(24, 10, 6, 3, 3, 8)$ and adjacency matrix

$$\begin{bmatrix} D & D & D & I & O & O \\ D & D & I & D & O & O \\ D & I & O & O & D & D \\ I & D & O & O & D & D \\ O & O & D & D & D & I \\ O & O & D & D & I & D \end{bmatrix},$$

where $D = J - I, J = J_4, I = I_4$ and $O = O_4$.

By replacing

$$D \to \begin{bmatrix} D & D & D & I \\ D & D & I & D \\ D & I & D & D \\ I & D & D & D \end{bmatrix}, \; I \to \begin{bmatrix} I & I & I & D \\ I & I & D & I \\ I & D & I & I \\ D & I & I & I \end{bmatrix}, \; O \to \begin{bmatrix} O & O & O & O \\ O & O & O & O \\ O & O & O & O \\ O & O & O & O \end{bmatrix}, \qquad (1)$$

we get a recursive construction for a DDG with parameters $(6{\cdot}4^t,\ 2{\cdot}4^t{+}2^t,\ 4^t + 2^t,\ 2 \cdot 4^{t-1} + 2^{t-1},\ 3,\ 2 \cdot 4^t)$. Denote by Γ^t this DDG.

Consider the subgraph formed by the first $2 \cdot 4^t$ vertices of Γ^t (in terms of Construction 1 this subgraph has adjacency matrix M) and the subgraph formed by the last $2 \cdot 4^t$ vertices of Γ^t (in terms of Construction 1 this subgraph has adjacency matrix N). Let Γ_1^t denote the first subgraph and Γ_2^t denote the second subgraph.

Common neighbours in Γ^t

Each vertex from Γ^t can be associated with the corresponding block-row of the adjacency block-matrix. Two blocks from distinct block-rows are called a pair if they lie in the same block-column.

Common neighbours in Γ^t

Lemma 8

Let x and y be two distinct vertices from Γ^t. The following cases hold.
(1) If the vertices x and y are associated with the same block-row, then, for this block-row and the vertices x and y, each block D gives 2 common neighbours and each block I or O gives 0 common neighbours.
(2) If the vertices x and y are associated with distinct block-rows, then the following subcases hold.
(2.1) If the vertices x and y correspond to the rows with the same index within their block-rows then, for these block-rows and the vertices x and y, each pair D and D gives 3 common neighbours; each pair I and I gives 1 common neighbour; each pair D and I or pair I and D gives 0 common neighbours; any pair with block O gives 0 common neighbours.

(2.2) If the vertices x and y correspond to rows with different indices within their block-rows, then, for these block-rows and the vertices x and y, each pair D and D gives 2 common neighbours; each pair I and I gives 0 common neighbours; each pair D and I or pair I and D gives 1 common neighbour; any pair with block O gives 0 common neighbours.

Blocks in the adjacency block-matrix

We investigate how the replacement of blocks in the recursive construction of Γ^t affects the adjacency block-matrix. In this investigation, we exclude all the cases related to the block O.

Lemma 9

For a block-row in the recursive construction of Γ^t, each block D gives 3 new blocks D and 1 new block I in each block-row of the adjacency block-matrix after replacing; each block I gives 1 new block D and 3 new blocks I in each block-row of the adjacency block-matrix after replacing.

Blocks in the adjacency block-matrix

Lemma 10

For two distinct block-rows in the recursive construction of Γ^t,the following cases hold.
(1) If two distinct block-rows are obtained from one block-row, then each block D gives 2 pairs D and D, 1 pair I and D and 1 pair D and I after replacing; each block I gives 2 pairs I and I, 1 pair I and D and 1 pair D and I after replacing.
(2) If two distinct block-rows are obtained from two distinct block-rows and have the same index within replaceable block-row, then each pair D and D gives 3 pairs D and D and 1 pair I and I; each pair I and I gives 3 pairs I and I and 1 pair D and D; each pair D and I gives 3 pairs D and I and 1 pair I and D; each pair I and D gives 3 pairs I and D and 1 pair D and I.

(3) If two distinct block-rows are obtained from two distinct block-rows and have different indices within replaceable block-row, then each pair D and D gives 2 pairs D and D, 1 pair D and I and 1 pair I and D; each pair I and I gives 2 pairs I and I, 1 pair D and I and 1 pair I and D; each pair D and I gives 3 pairs D and I and 1 pair I and D; each pair I and D gives 3 pairs I and D and 1 pair D and I.

Structural description of Γ_1^t and Γ_2^t

Lemma 11

The graph Γ_1^t is a DDG with parameters $(2 \cdot 4^t, 4^t + 2^t, 4^t + 2^t,\ 2 \cdot (4^{t-1} + 2^{t-1}),\ 4^t, 2)$.

Sketch of proof:

The graph Γ_1^1 has adjacency matrix $\begin{bmatrix} D & D \\ D & D \end{bmatrix}$.

For Γ_1^t, a recursive construction similar to (1) can be applied. Denote by A_1^t the adjacency block-matrix of Γ_1^t. Note that A_1^t does not contain a block O.

Structural description of Γ_1^t and Γ_2^t

The following table shows the number of blocks in each block-row of A_1^t according to Lemma 9.

t	D	I
1	2	0
2	6	2
3	20	12
$\dots$	$\dots$	$\dots$
t	$4^{t-1}+2^{t-1}$	$4^{t-1}-2^{t-1}$
$t+1$	4^t+2^t	4^t-2^t
$\dots$	$\dots$	$\dots$

Structural description of Γ_1^t and Γ_2^t

Note that, given a block-raw in A_1^t, there is exactly one block-row that is equal to the given one (that is, the block-row with the same sequence of blocks).

The next table shows the number of pairs of blocks in two distinct block-rows of A_1^t according to Lemma 10 (the case of equal block-rows is excluded, except for $t = 1$). Note that this table represents each of the cases from Lemma 10.

t	D, D	I, I	D, I	I, D
1	2	0	0	0
2	4	0	2	2
3	12	4	8	8
...	...	...	...	...
t	$2 \cdot (4^{t-2} + 2^{t-2})$	$2 \cdot (4^{t-2} - 2^{t-2})$	$2 \cdot 4^{t-2}$	$2 \cdot 4^{t-2}$
$t+1$	$2 \cdot (4^{t-1} + 2^{t-1})$	$2 \cdot (4^{t-1} - 2^{t-1})$	$2 \cdot 4^{t-1}$	$2 \cdot 4^{t-1}$
...	...	...	...	...

Structural description of Γ_1^t and Γ_2^t

After we apply Lemma 8 to count the number of common neighbours, we show that Γ_1^t is a DDG with parameters $(2 \cdot 4^t,\ 4^t + 2^t, 4^t + 2^t,\ 2 \cdot (4^{t-1} + 2^{t-1}),\ 4^t, 2)$. In particular, any pair of vertices corresponding to equal rows of the adjacency matrix forms a block of size 2 of the canonical partition.

Lemma 12

The graph Γ_2^t is a DDG with parameters $(2 \cdot 4^t, 4^t, 0,\ 2 \cdot 4^{t-1},\ 4^t, 2)$.

There are two known DDGs of diameter 2 with $\lambda_1 = 0$ (more generally, there are two known strictly Deza graphs with $a = 0$), one on 8 vertices and one on 32 vertices. The series Γ_2^t covers both cases and gives an infinite series of DDGs of diameter 2 with $\lambda_1 = 0$ (more generally, strictly Deza graphs with $a = 0$).

Vertex connectivity of Γ_2^t

Lemma 13

Let Γ be a connected Deza graph with parameters (v, k, b, a) with the second largest eigenvalue q. If Γ has a disconnecting set of minimum cardinality, that is not the neighbourhood of some vertex, then the following inequality holds: $k - 2q \leq b$.

Lemma 14

The vertex connectivity of Γ_2^t equals 4^t.

Sketch of proof:
Let us calculate the spectrum of Γ_2^t as a DDG:

$$\begin{aligned}\{k, \pm\sqrt{k-\lambda_1}, \pm\sqrt{k^2-\lambda_2 v}\} &= \{4^t, \pm\sqrt{4^t-0}, \pm\sqrt{(4^t)^2 - 2\cdot 4^{t-1}\cdot 2\cdot 4^t}\} \\ &= \{4^t, \pm 2^t, 0\}.\end{aligned}$$

The second largest eigenvalue of Γ_2^t is 2^t. Considering Γ_2^t as a Deza graph with the parameters (v, k, b, a), we get $b = 4^{t-1}$. So, the inequality from Lemma 13 becomes $4^t - 2\cdot 2^t \leq 4^{t-1}$, which holds only for $t = 1$. Thus, for any $t > 1$, the vertex connectivity of Γ_2^t equals k, where $k = 4^t$. If $t = 1$, then Γ_2^t is a DDG with parameters $(8, 4, 0, 2, 4, 2)$. By computer calculations using SageMath, the vertex connectivity of this graph equals 4.

Vertex connectivity of Γ^t

Lemma 15

The vertex connectivity of Γ^t equals $2 \cdot 4^t$.

Sketch of proof:

By Lemma 2, the vertex connectivity of Γ_1^t equals $4^t + 2^t$, By Lemma 14 the vertex connectivity of Γ_2^t equals 4^t. Thus, by Lemma 7, the vertex connectivity of Γ^t equals $2 \cdot 4^t$, which is 2^t less than the degree of a vertex.

Conclusion

There are more constructions of regular graphical Hadamard matrices with positive l, so in view of Construction 1, there are more DDGs whose vertex connectivity is less than k, where k is the degree of a vertex. In this talk we focused on the smallest graph from Construction 1 and its generalisation (in the sense of the recursive construction). We are interested if there exist examples of of DDGs, whose vertex connectivity is less than k.

Conclusion

There are some approaches for obtaining more general results about the vertex connectivity of DDGs. For example, the approach that was used in [6] and [8], or the approach that was used in [7]. Since the spectrum of a DDG is not completely determined by its parameters, the first approach can only be used in specific cases like Lemma 14. The second approach requires more detailed consideration and possible development of new tools to apply it to DDGs. We are interested if general results will be obtained for DDGs, using both known approaches.

Thanks for your attention!